Network Analysis from Scratch: Introduction to Gephi
Session 2 Handout

Brian Tsz Ho Wong, Ph.D.
Training Fellow, Edinburgh Futures Institute, University of Edinburgh
Email: twong4@ed.ac.uk

14 October 2026

We will use Harvard University's China Biographical Database Project (CBDB) to demonstrate Gephi's GeoLayout and ExportToEarth features. CBDB is one of the largest and most comprehensive biographical databases for the study of Chinese history. It covers Chinese historical figures from the last 1000 years.

The CBDB has many features, we will not go through them all. However, if you are interested, there is a very detailed user guide on the project's website (https://cbdb.hsites.harvard.edu/sites/g/files/omnuum3101/files/cbdb/files/cbdb_users_guide.pdf). The user's guide is 153 pages long, you don't need to (and don't waste your time!) to read it.

In class we will use the example of the Xiang Army (*Xiang jun*, also known as the Hunan Army) coalition. We will look at the social networks of some of its key leaders, including Luo Zenan (1807-1856), Zeng Guofan (1811-1872), Hu Linyi (1812-1861), Zuo Zongtang (1812-1885), and Peng Yulin (1816-1890). The Xiang Army was organised by Zeng Guofan and used by the Qing government to suppress the Taiping Rebellion in the mid-19th century. The leaders of the Xiang Army, such as Zeng and his brothers, disciples, and allies, became statesmen and dominated the Qing court in the late 19th and early 20th centuries. Using Gephi and its GeoLayout and ExportToEarth functions, we will discover the effectiveness and limitations of CBDB.

1.1. Import a CBDB network into Gephi

1. IntroNetworkAnalysis repository on github > GephiFiles.
2. Download Hu Linyi.gdf, LuoZenan.gdf, PengYulin.gdf, Zeng Guofan.gdf, and Zuo Zongtang.gdf.
3. Open Gephi.
4. Click on 'Open Graph File' and then select the Zeng Guofan.gdf and press 'open'.
5. Select 'Geo Layout' for the layout.

6. Go to ‘Data Laboratory’, and remove all the nodes with a value of 0 in the latitude and longitude columns.
7. Add two columns: ‘lat’ and ‘lng’. Set both columns to ‘double’.
8. Select ‘Copy data to other column’, copy the values in the ‘latitude’ column to the ‘lat’ column, and then copy the values in the ‘longitude’ column to the ‘lng’ column.
9. Go to ‘Overview’, select ‘Map of Countries’ for the layout.
10. Under ‘Layout’, select ‘Geo Layout’. Ensure that ‘lat’ is selected for ‘latitude’ and ‘lng’ for ‘longitude’.
11. Click ‘Run’

1.2. Export a CBDB network to Google Earth (*To use this feature, you will need an older version of Gephi, such as version 0.9.3. This is not possible on the latest version of Gephi. You can download the older version from the Gephi website, https://github.com/gephi/gephi/releases.**)**

1. File > Export > Graph File
2. Click on ‘Options’, and make sure that ‘lat’ is selected for ‘latitude’ and ‘lng’ for ‘longitude’.
3. Export the network in a KMZ file

Exercise 1: Do the same for Hu Linyi.gdf, LuoZenan.gdf, PengYulin.gdf, and Zuo Zongtang.gdf.

2.1. What is missing?

1. Go to https://histecon.fas.harvard.edu/visualizing/graphing/geography.html to download the nodes data and edges data for this project.
2. Discuss with your group what data on nodes and edges is missing in order to create a Gephi dataset.

2.2. Create a Gephi dataset with coordinates

1. For the ‘nodes data’, copy rows 479–593 (locations and institutions) to a new spreadsheet.
2. Remove rows 479-593. Then save the file.
3. Open OpenRefine.
4. Upload the ‘nodes data’ > Create Project.
5. All > Edit columns > Reorder/remove columns > Keep only the ‘Label’ and ‘Birth Nation’ columns.

6. Label > Edit cells > Blank down.
7. Label > Facet > Custom text facet.
8. Enter the GREL: isBlank(cells["Label"].value).
9. Go to Facet/Filter.
10. Choose ‘False’.
11. Birth Nation > Edit cells > Replace.
12. Enter ‘?’ for the ‘Find’ box and leave the ‘Replace with’ box blank.
13. Birth Nation > Facet > Custom text facet.
14. Enter the GREL: isBlank(cells["Birth Nation"].value).
15. Go to Facet/Filter.
16. Choose ‘False’.
17. Export the table as an excel file.
18. Open the table and add the ‘lat’ and ‘lng’ columns.
19. Open any generative model, copy the table into it, and then enter the following prompt: Add the coordinates for all ‘Birth Nation’ entries in the table.
20. Copy the table to a new spreadsheet.
21. Copy rows 479–593 and paste them under the name list.
22. Add an row ‘ID’ to the left of the ‘Label’ column. Give an ID to each entity.
23. Save the spreadsheet as the ‘Nodes’ table.
24. For the ‘edges data’, remove rows 479-593.
25. Remove columns E-J.
26. Enter ‘1’ for each entry in the ‘weight’ column.
27. Remove the empty rows.
28. Add a column between the ‘Sourcename’ and ‘Targetname’ columns, and another between the ‘Targetname’ and ‘weight’ columns. Name these columns ‘Source’ and ‘Target’, respectively.
29. Copy the ‘ID’ and ‘Label’ columns from the ‘Nodes’ table to the ‘InstitutionsGEO Edges v.2’ spreadsheet.
30. Use the xlookup function to find the IDs for the entries of the ‘Sourcename’ and ‘Targetname’ columns.
31. To remove the ‘#N/A’ values in the ‘Target’ columns, we can either remove it manually or do this in OpenRefine.
32. Open OpenRefine.
33. Upload the ‘edges data’ > Create Project.
34. All > Edit columns > Reorder/remove columns > Keep the ‘Source’, ‘Target‘, ‘Type’, and ‘weights’ columns.
35. Target > Edit cells > Replace.
36. Enter ‘#N/A’ for the ‘Find’ box and leave the ‘Replace with’ box blank.
37. Target > Facet > Custom text facet.
38. Enter the GREL: isBlank(cells["Target"].value).

39. Go to Facet/Filter.
40. Choose ‘False’.
41. Export the table as an excel file.
42. Rename the spreadsheet ‘Edges’.
43. Combine the ‘Nodes’ and ‘Edges’ tables into one Excel file. Rename the file ‘Cambridge Economists’.
44. Import the excel file into Gephi.
45. For the ‘Nodes’ table, select ‘double’ for ‘lat’ and ‘lng’.
46. For the ‘Edges’ table, select ‘Integer’ for ‘weight’.
47. Select ‘Geo Layout’ for the layout.
48. Select ‘Map of Countries’ for the layout.
49. Under ‘Layout’, select ‘Geo Layout’. Ensure that ‘lat’ is selected for ‘latitude’ and ‘lng’ for ‘longitude’. Click ‘Run’

Exercise two: Repeat the aforementioned steps to create a Gephi dataset with coordinates. The data can be downloaded here:
https://histecon.fas.harvard.edu/visualizing/graphing/geography.html